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Location of the Lhasa terrane in the Late Cretaceous and its implications for crustal deformation

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ABSTRACT

The Cenozoic crustal deformation of the Asian continent is constrained by the location of its southern margin in the Late Cretaceous, which is, however, affected by potential remagnetization and inclination shallowing of redbeds in the Linzhou Basin. Our paleomagnetic results from the Shexing Formation redbeds in the Late Cretaceous yield a mean paleo-inclination of ~21.5°, consistent with the recalculation of previous data. Rock-magnetic, paleomagnetic and petrographic analyses indicate that the redbeds have recorded the original paleomagnetic field and have not suffered from significant inclination shallowing. The paleopole from 70 redbeds sites is at 71.8°N, 281.5°E with $A_{95} = 2.6°$, yielding a paleolatitude of ~12.0 ± 2.6°N at ~70–91 Ma for the Linzhou Basin. This result indicates a latitudinal convergence of 1264 ± 477 km north of the sampling area within Asia since the Late Cretaceous. Combined with previous data, our results show that the southern margin of Asia was at ~11–15°N in the Late Cretaceous, aligning in a WNW-ESE direction, which is similar to its present-day orientation. A review of available data shows that the sharp bending of the southern margin of Asia around the eastern Himalayan syntaxis (EHS) is probably due to different crustal movement patterns around the EHS. The shape of the southern margin of Asia determines the arc-shaped orientation of Himalayan orogen.

1. Introduction

During the Cenozoic, the India-Asia collision created the Himalayan-Tibetan Orogen, resulting in the closure of the Neo-Tethys Ocean, substantial crustal shortening, and the southeastward extrusion of the Tibetan Plateau (van Hinsbergen et al., 2011; Yin, 2010). The Neo-Tethys Ocean's closure is represented by the Indus-Tsangpo suture zone (Metcalfe, 2013), which separated the Lhasa terrane to the north and from the Himalaya to the south (Fig. 1a and b). Notably, to the east of the eastern Himalayan syntaxis (EHS), both Neo- and Meso-Tethys sutures and their eastward extension bend sharply to an N-S direction (Fig. 1a).

The Lhasa terrane was the southern margin of the Asian continent before its collision with India (Yin and Harrison, 2000; van Hinsbergen et al., 2012). The present-day elongation of Lhasa terrane is NWW-SEE, while its shape before the India-Asia collision remained controversial. Paleogeographic reconstruction of the Lhasa terrane can help us better understand this area's tectonic evolution and deformation mechanism. The shape of the pre-collisional southern margin of Asia is critical in understanding the onset of the India-Asia collision (e.g., Lippert et al., 2014; Yi et al., 2011, 2015), as well as the amount of crustal shortening within the Asian continent (e.g., Lippert et al., 2014; Yi et al., 2015). Because the crustal shortening is a fundamental cause for the uplift of a plateau (e.g., Wang et al., 2008; Li et al., 2015; Meng et al., 2017; Murphy et al., 1997), the magnitude and timing of crustal shortening have a close relation with the uplift history of the Tibetan Plateau (Meng et al., 2017; Tong et al., 2016, 2017a). Therefore, a precise paleogeographic reconstruction and paleolatitude evolution of the Lhasa terrane can better constrain the magnitude and timing of crustal shortening and

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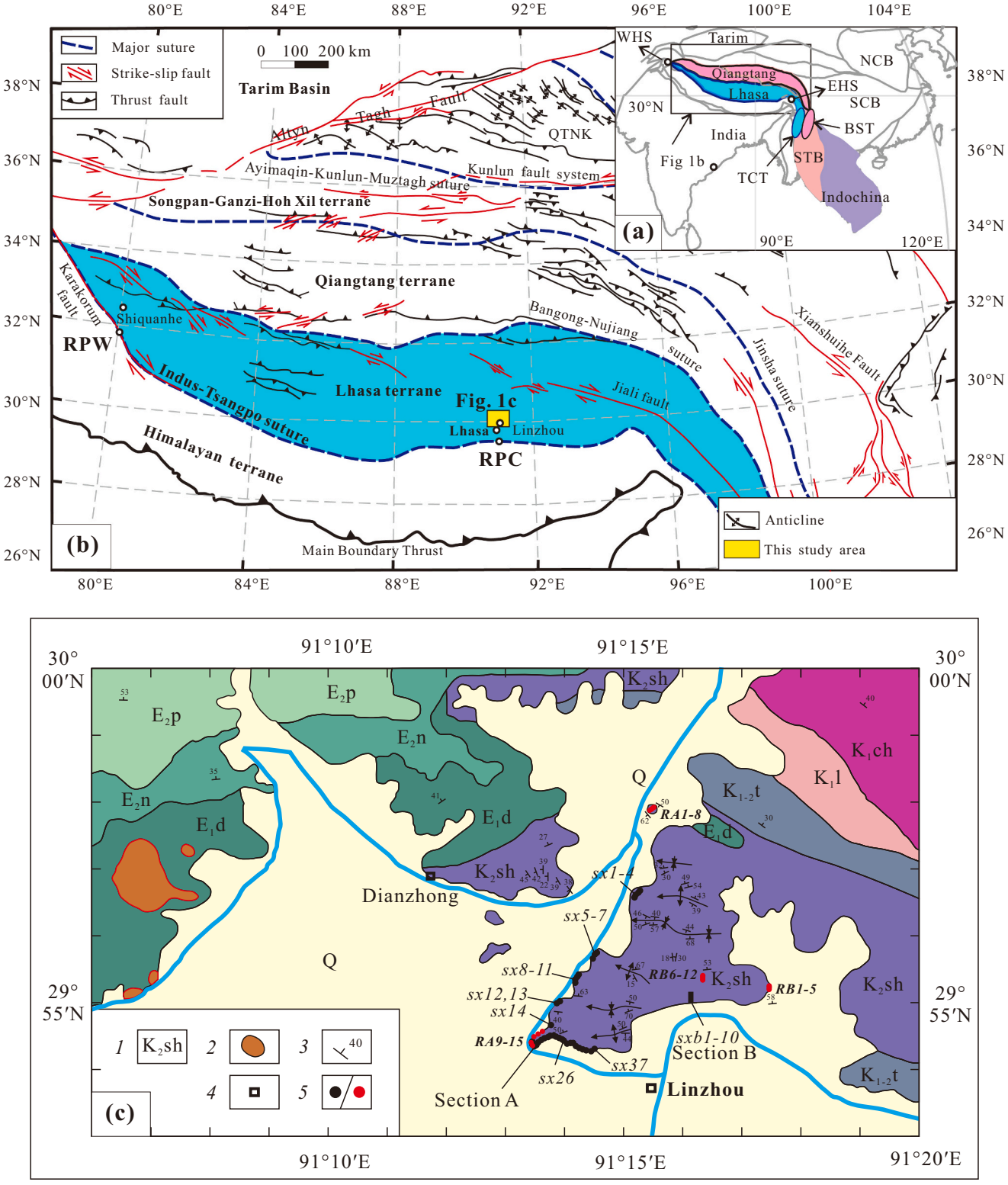


Fig. 1. (a, b) Regional geological map of the Tibetan Plateau and adjacent areas, modified from Yin and Harrison (2000) and Li et al. (2015). Abbreviations: BST, Baoshan terrane; NCB, North China block; QTNK, Qimen Tagh-North Kunlun thrust system; SCB, South China block; STB, Shan-Thai block; TCT, Tengchong terrane; EHS, eastern Himalayan syntaxis; WHS, western Himalayan syntaxis. (c) Geological map of the sampling area, modified from Tan et al. (2010) based on our observation. 1, stratigraphic units including P_2p (Upper Permian Pangna Fm), K_1l (Lower Cretaceous Linbuzong Fm), K_1ch (Lower Cretaceous Chumulong Fm), $K_{1\text{-}2}t$ (Cretaceous Takena Fm), K_2sh (Upper Cretaceous Shexing Fm), E_1d (Paleocene Dianzhong Fm), E_2n (Eocene Nianbo Fm), E_2p (Eocene Pana Fm) and Q (Quaternary). 2, intrusive units. 3, strike and dip of bedding. 4, town. 5, Sampling locations of Tan et al. (2010) (black dots) and this study (red dots). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

plateau uplift since the collision.

Although numerous geological and geophysical investigations have been carried out in the Lhasa terrane in the last few decades, its paleogeographic location is still poorly constrained. Paleomagnetism has been widely used in reconstructing the paleogeography of the southern margin of Asia. Unlike the present-day NWW-SEE orientation, the original southern margin of Asia was mainly rebuilt into two contrasting orientations, namely a relatively *E*-W direction (e.g., Ma et al., 2014) and a relatively linear structure with an NW-SE orientation (e.g., Yi et al., 2015). Yi et al. (2015) restored the southern margin of Asia into a relatively linear structure with an orientation of NW 310° by comparing the paleolatitude of the western part of Lhasa terrane, which is observed from the Cretaceous volcanic and sedimentary rocks, with that of the central part of Lhasa terrane, which is derived from the Paleocene-Eocene Linzizong volcanics (e.g., Chen et al., 2014). Given that the Linzizong volcanics erupted at ~40–69 Ma (He et al., 2007; Tan et al., 2010; Huang et al., 2013), which may have erupted after the India-Asia collision, they thus cannot provide a critical location constraint for the pre-collision southern margin of Asia (e.g., Lippert et al., 2014; Yi et al., 2015). Therefore, the Late Cretaceous paleomagnetic results are vital to recovering the pre-collisional paleolatitudes of Lhasa terrane at different localities. A series of paleomagnetic investigations on different rock types have shown that the western part of the Lhasa terrane was located at a stable paleolatitude of ~14–20°N during the Cretaceous (e.g., Chen et al., 2012; Ma et al., 2014, 2017a, 2017b, 2018; Bian et al., 2017, 2020; Yang et al., 2015a; Yi et al., 2015). For the central and eastern

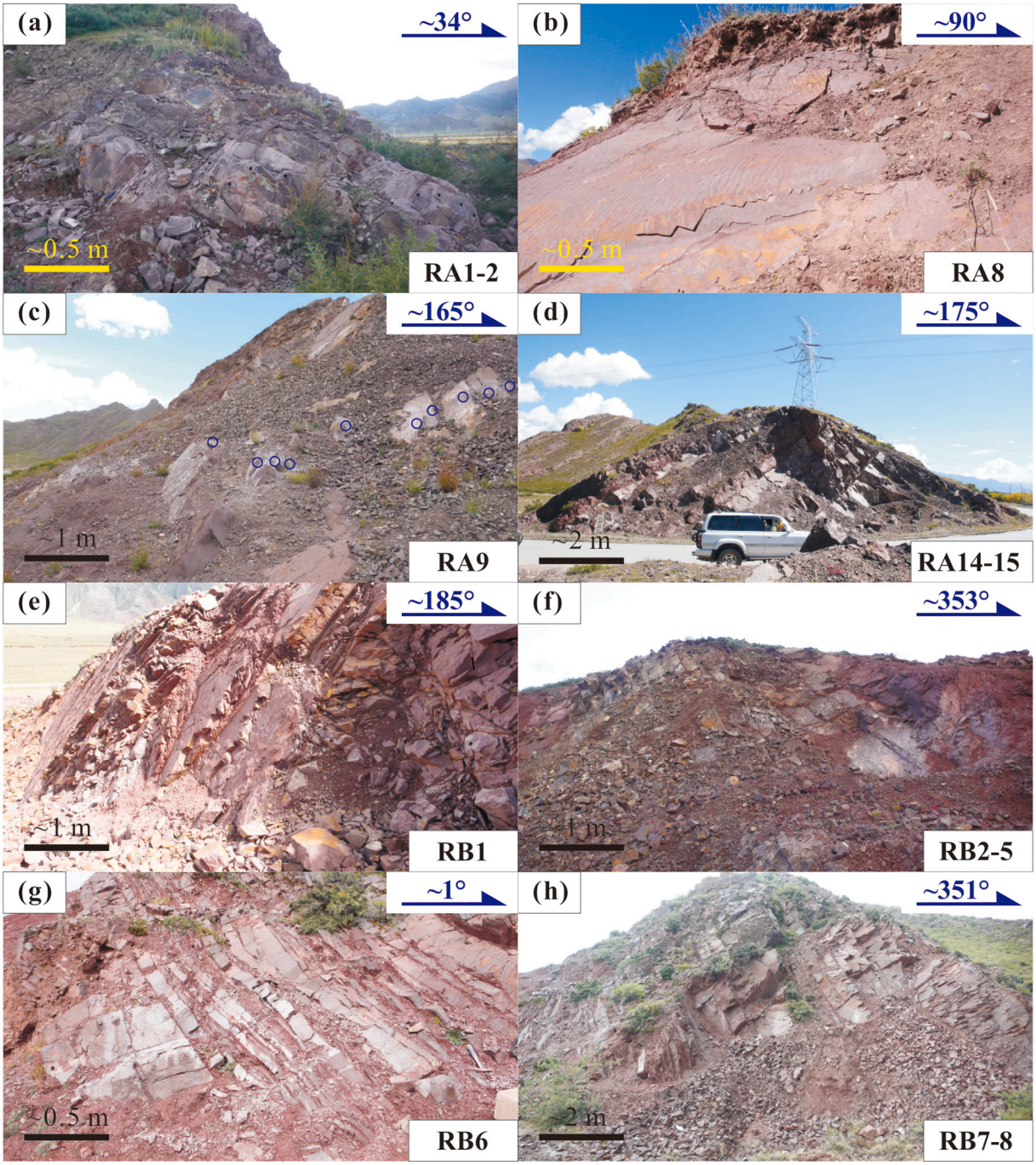


Fig. 2. Field outcrops of sampling sections and sites of the Shexing Fm redbeds showing explicit bedding attitudes. (a, b), (c, d), (e, f), and (g, h) are four different outcrops.

parts of the Lhasa terrane, most paleomagnetic investigations, focusing on the redbeds, yield paleolatitudes ranging from ~10°N to ~24°N (Tan et al., 2010; Sun et al., 2012; van Hinsbergen et al., 2012; Cao et al., 2017; Liang et al., 2017; Tong et al., 2017b).

Some researchers suggested that the redbeds from Linzhou Basin have suffered from significant inclination shallowing based on elongation/inclination (E/I) correction (Tan et al., 2010; van Hinsbergen et al., 2012), while others suggested that the inclination shallowing is insignificant (Cao et al., 2017; Sun et al., 2012; Tong et al., 2017b). Furthermore, some studies revealed that the redbeds in Linzhou Basin might have suffered from remagnetization based on the *syn*-folding test and the AMS and petrological investigations (Liang et al., 2017). Considering that previous paleomagnetic studies on the redbeds from the central part of Lhasa terrane may suffer from potential remagnetization or inclination shallowing, the redbeds from the Linzhou area are reinvestigated with rock-magnetic, petrographical, and paleomagnetic analyses to reconstruct the southern margin of Lhasa terrane and to deduce its tectonic implications.

2. Geology and sampling

Geological and paleomagnetic observations have suggested that the age of Lhasa-Qiangtang collision ranges from Middle Jurassic to Late Cretaceous (e.g. Yin and Harrison, 2000; Kapp et al., 2007; Zhang et al., 2012; Zhu et al., 2016; Ma et al., 2017a; Ma et al., 2018; Sun et al., 2019; Cao et al., 2020). The Lhasa terrane underwent significant crustal shortening during the Cretaceous (e.g., Kapp et al., 2005; van Hinsbergen et al., 2011), partly absorbed by intensely folded Cretaceous strata. The Cretaceous strata include Shexing, Takena, Chumulong, and Linbuzong Formations (Fms). The Shexing Fm is sandwiched unconformably between the overlying Dianzhong Fm of Linzizong Group and the underlying Takena Fm (Chen, 2017). The angular unconformities between Shexing Fm and Dianzhong Fm or Takena Fm have been documented in previous studies (Chen, 2017; He et al., 2007) (see cross-sections in Fig. 3-2 and 3-6 of Chen, 2017). Previous studies suggested that the Linzizong Group is formed at ~40–69 Ma (Tan et al., 2010; He et al., 2007). Recently, the volcanic rocks, which angularly overlie the Shexing Fm, were dated to be 69.7 ± 0.3 Ma (Chen, 2017). The angular unconformity between them indicates that the Shexing Fm strata fold is formed earlier than ~70 Ma. The youngest detrital zircon grains from the uppermost part of the Takena Fm yield a weighted mean $^{206}Pb/^{238}U$ age of ~95 Ma, revealing a maximum depositional age of ~95 Ma for the uppermost part of the Takena Fm (Chen, 2017). Together with the dikes intruded in the upper part of the Takena Fm at ~91 Ma, these results indicate that the uppermost part of the Takena Fm was deposited at ~91–95 Ma (Chen, 2017). Therefore, Shexing Fm redbeds were likely deposited during the interval from ~91 to ~70 Ma. This conclusion can be further supported by four interbedded volcanic rocks of the Shexing Fm dated at ~85, ~82, ~79 and ~ 78 Ma (Chen, 2017). Notably, Wang et al. (2020) conclude that the Shexing Fm is deposited during ~96–119 Ma, or even down to ~90 Ma based on the detrital-zircon chronostratigraphy of the sedimentary rocks from the Shexing Formation. Because the deposition age of the Shexing Fm strata should be younger than the youngest detrital zircon ages revealed by Wang et al. (2020), the disparate zircon ages from Wang et al. (2020) and Chen (2017) are not paradoxical. In addition, Cao et al. (2017) reported paleomagnetic results of the Shexing Fm with several reversals. Therefore, a deposition age of Cretaceous Normal Superchron (~83.6–121.0 Ma) for the Shexing Fm could be excluded (Gradstein et al., 2020). We suggest that the Shexing Fm redbeds should be deposited at ~70–91 Ma.

To test whether the redbeds from the Linzhou area have suffered from remagnetization, we collected 27 sites of Shexing Fm redbeds from four outcrops in the Linzhou area (Fig. 1c). Sites RA1–RA8, RA9–RA15, RB1–RB5, and RB6–RB12 were collected from 4 different outcrops with a straightforward bedding attitude (Figs. 1c and 2). A previous paleomagnetic study has collected 47 sites redbeds, i.e., sites sx1-sx37 and sxb1-sxb10, in the Linzhou Basin (Fig. 1c) (Tan et al., 2010). Sites RA9–RA15 in this study were collected from previous sampling sections (Fig. 1c). The other three sections were collected from new sampling locations. Several anticlines and synclines can be found in the Linzhou area (He et al., 2007). The asymmetric bedding attitudes of different sampling sections do not share the same plunging fold, as shown in Fig. 1c. Each site contains at least eight cores oriented with magnetic and Sun compasses, indicating no local magnetic disturbance in our sampling sections. Six oriented block samples named after their corresponding paleomagnetic sites were collected to evaluate the influence of possible inclination shallowing.

3. Experimental and analytical method

Cores were drilled perpendicularly to the strata bedding in six hand samples in the laboratory. All cores were cut into 2.2-cm-long specimens, and fresh off-cuts were collected for rock magnetic investigations. The acquisition of isothermal remanent magnetization (IRM) and backfield demagnetization of saturation IRM (SIRM) of representative specimens were performed with an ASC IM10–30 pulse magnetizer. IRM was measured using a JR-6A spinner magnetometer at Guangzhou Institute of Geochemistry, Chinese Academy of Sciences. The thermal demagnetization of a three-component IRM was performed with an ASC-TD 48 furnace and a JR-6A spinner magnetometer in the Paleomagnetic and Environmental Magnetism Laboratory (PMEML) at China University of Geosciences, Beijing (CUGB) (Lowrie, 1990). Hysteresis loops and magnetic hysteresis parameters of typical specimens were measured on a MicroMag 3900 vibrating sample magnetometer in the Paleomagnetism and Geochronology Laboratory at Institute of Geology and Geophysics, Chinese Academy of Sciences. Most specimens were treated with stepwise thermal demagnetization from 100 to ~680 °C, and the remanence was measured with a 2G 3-axis cryogenic magnetometer in a shielded room with residual fields less than 300 nT in PMEML at CUGB. The AMS of three or more random specimens from each site were measured using an AGICO KLY-4 Kappabridge in PMEML.

Notably, Hodych and Buchan (1994) proposed an anisotropy-based method for inclination shallowing correction. This method has been widely used to quantitatively evaluate the extent of inclination deviation (Tong et al., 2017b; Yang et al., 2019; Bian et al., 2020; Yuan et al., 2020). The oriented specimens were subjected to successively increasing direct-current fields (in the range of 0–1000 mT) applied at a 45° angle to the strata bedding plane. The IRM was measured after each application of the direct-current field, and the specimens were then treated with stepwise thermal demagnetization as natural samples. The IRMz and IRMx can be easily measured by using a JR-6A spinner magnetometer.

The characteristic remanent magnetization (ChRM) directions of specimens were determined and analyzed using principal component analysis (Kirschvink, 1980). Site-mean directions were determined using Fisherian statistics (Fisher, 1953). Paleomagnetic data were analyzed using computer program packages developed by Enkin (1990) and Cogné (2003).

Raman micro-spectroscopy was conducted at the State Key Laboratory of Biogeology and Environmental Geology, China University of Geosciences, Wuhan, using a WITec α 300 Confocal Raman system coupled with a Peltier cooled EMCCD detector.

4. Results

4.1. Rock-magnetic results

The IRM curves show a slow increase below 200 mT and are far from saturation even at 1.5 T (Fig. 3a-3d). Combined with the maximum remanence coercivity of ~395–634 mT (Fig. 3a-3d), these results reveal the high coercivity of magnetic carriers. The unblocked temperature at ~680 °C presented in the Lowrie test shows that the high coercivity

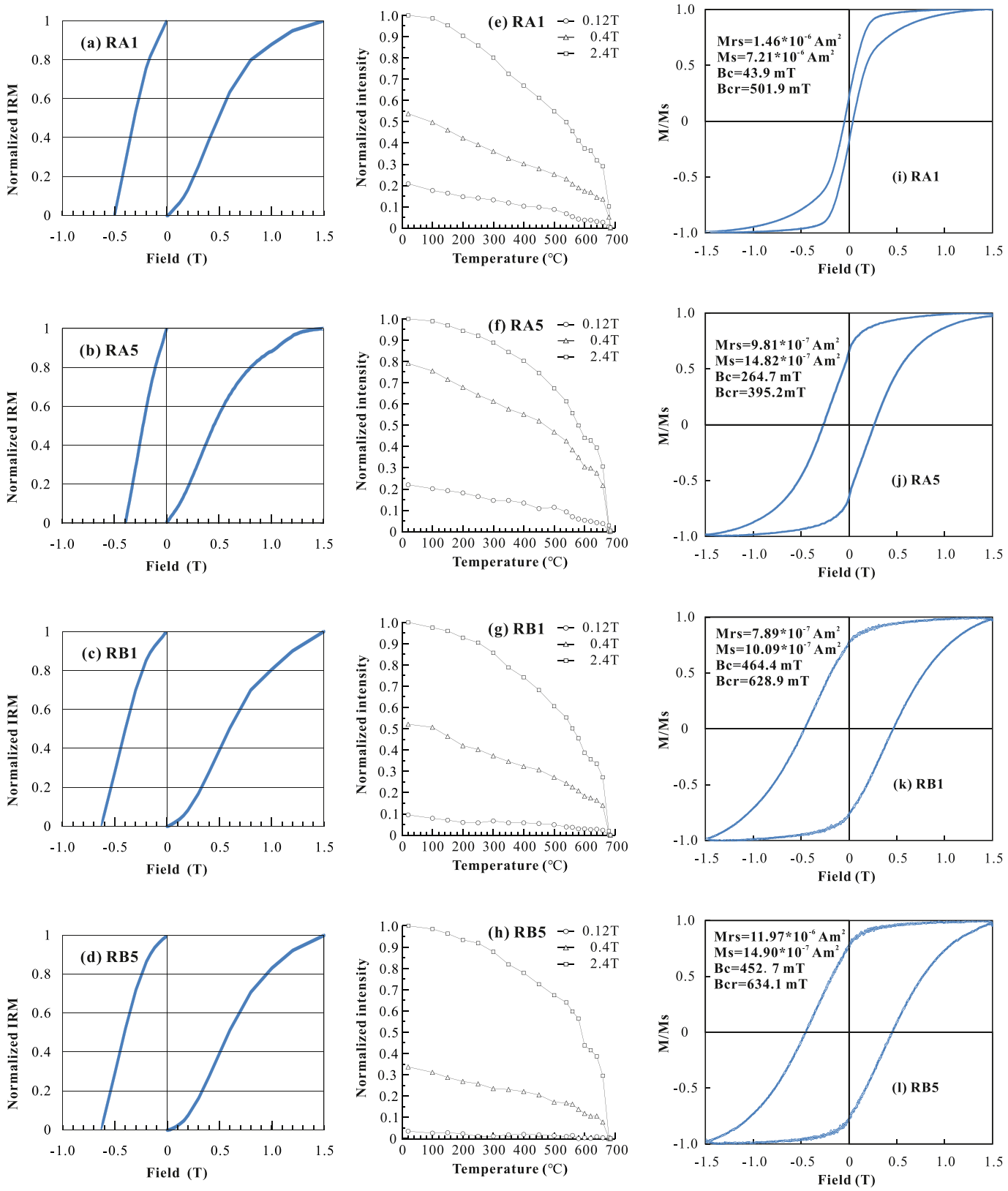


Fig. 3. IRM acquisition curves and back-field demagnetization of the SIRM curves (a-d), thermal demagnetization of a three-component IRM (e-h), and the hysteresis loops (i-l) of typical specimens (Lowrie, 1990). Mrs. is saturation remanent magnetization; Ms. is saturation magnetization; Bc is coercivity; Bcr is coercivity of remanence.

magnetic carriers are hematite (Fig. 3e-3h). The significant decrease of remanence intensity at ~580 °C and an unblocking temperature of ~580 °C for the soft component in the sample RB5 together indicate the existence of minor magnetite (Fig. 3e-3h). These interpretations are further supported by the wide or "wasp-waisted" hysteresis loops (Fig. 3i-3l). Therefore, we conclude that the primary magnetic carriers are hematite with minor magnetite.

Mean magnetic susceptibility (*K*m) from four sections varies from 79 $\times 10^{-6}$ to 151 $\times 10^{-6}$ (<500μSI). Because the volume susceptibility of magnetite is ~1000 times greater than that of most minerals (Gilder et al., 2001), the weak Km suggests that the AMS is dominantly controlled by paramagnetic minerals (mainly phyllosilicates) and antiferromagnetic minerals (hematite) (Huang et al., 2006; Weil and Yonkee, 2009; Li et al., 2020a). This conclusion is consistent with the Lowrie experiments, implying that the magnetite volume is insufficient and that the hematite is the dominant magnetic mineral in our redbed samples. The AMS results of four different sections are shown in Fig. 4. Four mean anisotropy degrees (Pj) of 1.014 ± 0.004, 1.031 ± 0.005, 1.024 ± 0.006, 1.022 ± 0.007 are revealed for sections RA1–8, RA9–15, RB1–5 and RB6–12 of the Shexing Fm redbeds, respectively (Fig. 4). The mean anisotropy degree (Pj) of all measured specimens is 1.023 ± 0.008. The mean magnetic foliation (1.014) is greater than the mean magnetic lineation (1.008), and both are extremely small.

The minimum AMS principal axes are either subparallel or inclined with a moderate to steep angle to the bedding plane of the measured specimens and are distributed in an approximately N-S orientation. These AMS structures, which are repetitively observed from the Shexing Fm in Linzhou Basin and its surrounding areas (Liang et al., 2017), are consistent with the N-S shortening. The folds with W-E hinge in the sampling areas indicate that the strata should have been affected by tectonic strain. Based on the distributions of the AMS principal axes and the classification of AMS structure proposed by Hrouda and Chadima, 2020, the deformation characteristics of our sampling redbeds should belong to stages 2 (incipient deformation) to 4 (weak cleavage), suggesting that the redbeds may have experienced weak to moderate compaction or tectonic stress (Fig. 4). This conclusion is consistent with that the finding that soft sedimentary deformation could not be found in the field based on the criteria for the recognition of soft sedimentary deformation (Alsop et al., 2019), and that the bedding attitudes of our sampling sections are evident in different scales from outcrop in the field to sampling cores.

4.2. Paleomagnetic results

Thermal demagnetization diagrams of the representative Shexing Fm redbeds specimens are shown in Fig. 5a-5r. Most specimens have a low-temperature component. The low-temperature components may be a remagnetization at the time interval between Late Cretaceous and present. Therefore, some low-temperature components are distributed around the present-day geomagnetic field direction (Dec = 0.0°, Inc. = 45.7°). Some specimens show scattered demagnetization at a high temperature near 680 °C and cannot fully decay to the origin. Nevertheless, a high-temperature component (HTC) toward the origin carried by detrital hematite can be determined from ~600 to ~680 °C for most specimens (Fig. 5a-5r).

HTC directions of 208 specimens from 27 sites are listed in Table S1 (supporting information). The site-mean directions are shown in the equal-area projections. The Fisher mean HTC directions of the 27 sites are Dg = 15.7°, Ig = 46.5°, kg = 4.6, and α_{95} = 14.6° in situ and Ds = 3.4°, Is = 21.5°, ks = 25.8, and α_{95} = 5.6° after tilt correction (Fig. 6a and b; Table 1). The arc stretching of data in Fig. 6b may result from potential vertical-axis rotations or plunging folds. Local rotations and plunging folds are very common, especially in the orogenic belt like Tibetan Plateau (e.g., Ma et al., 2014; Song et al., 2017; Yang et al., 2019). To evaluate the potential influence of the arc stretching of data on inclination mean, we calculated and obtained the mean inclination of 20.9 ± 3.3° for the 27 sites' redbeds using the inclination-only method (Arason and Levi, 2010). The mean value (20.9 ± 3.3°) of the inclination-only is well consistent with the Fisher mean inclination (21.5 ± 5.6°). Therefore, we conclude that the arc stretching of data in Fig. 6b has no significant influence on the inclination of the Fisher mean. Because of the Ks (25.8) / Kg (4.6) ratio of 5.6, the site-mean HTC directions show significant convergence after till correction. The McFadden (1990) correlation fold tests yield $\xi_{1,g} = 21.07$ and $\xi_{1,s} = 7.00$ in the geographic and stratigraphic coordinates, respectively. The critical value of the test statistic ξ is 8.55 at 99% confidence levels, revealing a positive fold test. The HTC directions yield a maximum grouping at 85% unfolding (Fig. 6c). The mean direction at 85% unfolding is not significantly different from that of the full-tilt correction at the 95% confidence level (Fig. 6b and c). These results indicate that the HTC directions have a prefold origin. Given that the folding occurred before ~70 Ma, the HTC directions likely represent the primary remanence.

4.3. Magnetic petrography

The petrographic thin sections of typical redbed specimen show a clastic structure (Fig. 7a). Detrital hematite grains and pigmentary hematite are abundant in the particle-supported redbeds (Fig. 7b-7e). The shapes of detrital hematite grains range from angular to subrounded at high illumination in the particle-supported siltstone (Fig. 7b, c), while pigmentary hematite occurs as the rims of clasts (Fig. 7c-e) (Liang et al., 2017), responsible for the color of the redbeds. Both detrital and pigmentary hematites can be identified based on their characteristic spectra of Raman micro-spectroscopy analyses (Fig. 7e-g). The pigmentary hematite may record a chemical remanent magnetization (CRM) during post-depositional chemical processes (Jiang et al., 2015; Liang et al., 2017; Meng et al., 2018). A similar characteristic has been described for the redbeds in the Tibetan Plateau (Liang et al., 2017; Meng et al., 2018). The detrital hematite grains most likely record a detrital remanent magnetization (DRM) during depositional and compaction processes. Lack of significant compactional signature in the minerals of the redbeds implies no significant inclination shallowing.

4.4. Corrections for inclination shallowing

The plots of IRMz, IRMx, and IRMz/IRMx of the Shexing Fm redbeds are shown in Fig. 8. The IRMz/IRMx values at 200–1000 mT and 600–685 °C vary from 0.91 to 1.05 and from 0.88 to 1.08, respectively. The HTC was determined in the range of ~600–680 °C, and the mean IRMz/IRMx ratio of 0.97 is roughly equivalent to 1, indicating that the inclination shallowing is negligible. Given that the anisotropy of the unique axis IRM method may undervalue the influence of inclination shallowing in some relatively coarse-grained redbeds (Tan et al., 2003), we carefully check the petrography of detrital hematite grains and other minerals and find no significant compactional signature (Meng et al., 2018).

5. Discussion

5.1. Did the redbeds from the Linzhou area record a primary remanence?

A critical aspect of paleomagnetic investigations on redbeds is to distinguish CRM from DRM. CRM is carried by fine-grained hematite pigment (Collinson, 1966; Tauxe et al., 1980), while DRM is recorded by detrital hematite particles with larger grain sizes (Collinson, 1965; Tauxe and Kent, 1984). The differences in crystallinity and grain size lead to distinctions in thermal decay curves: a concave shape with unblocking from ~200 to ~650 °C for CRM and a convex shape with unblocking from ~600 to ~680 °C for DRM (Jiang et al., 2015), suggesting that the shape of thermal decay curve can be used as a discriminator to distinguish these two remanences. The unblocking temperature at ~660–680 °C and the convex shape of the thermal decay

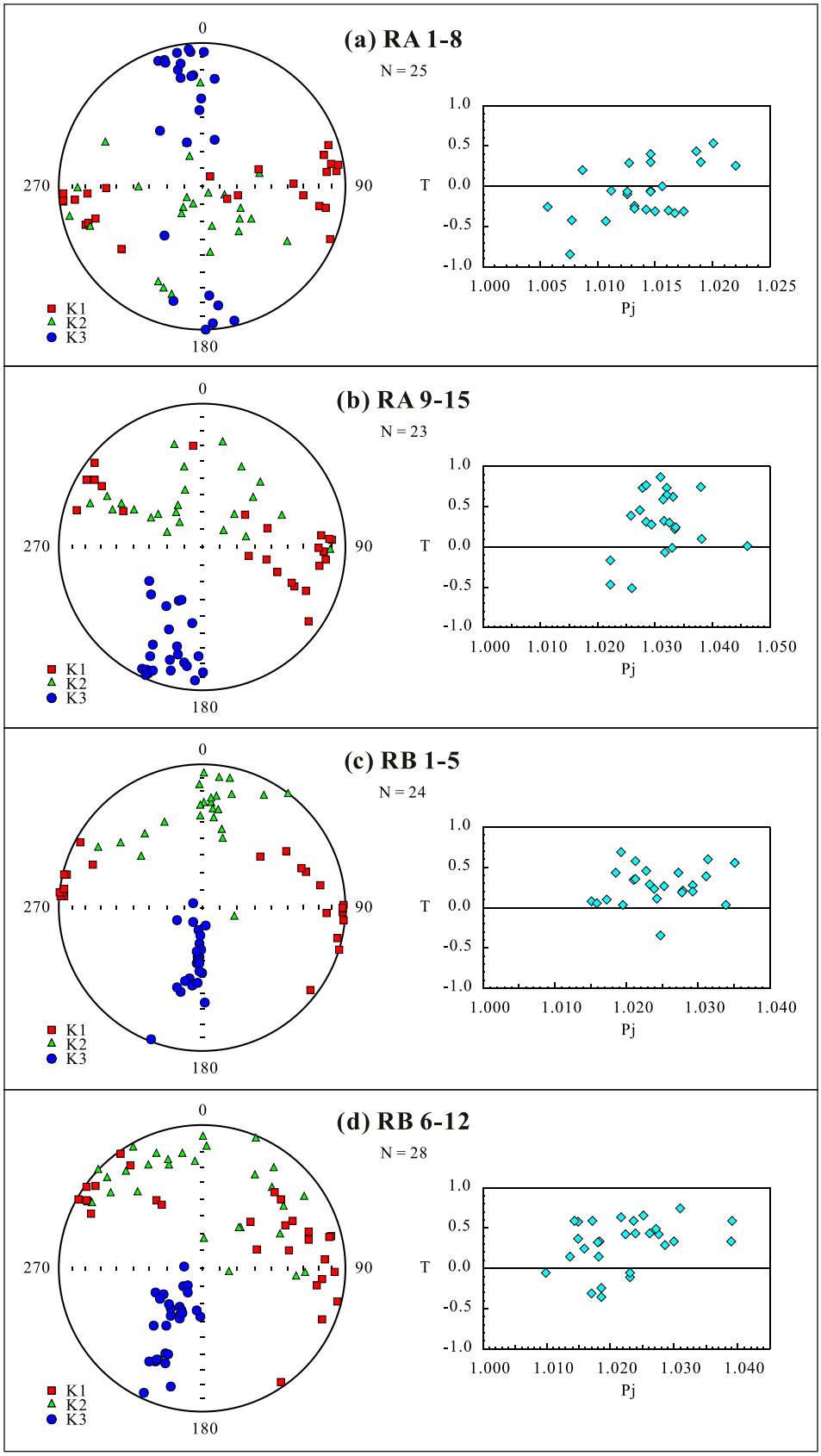


Fig. 4. Anisotropy of magnetic susceptibility data in stratigraphic coordinates and plots of shape parameter (T) versus anisotropy degree (Pj) of the Shexing formation redbeds from four different sections.

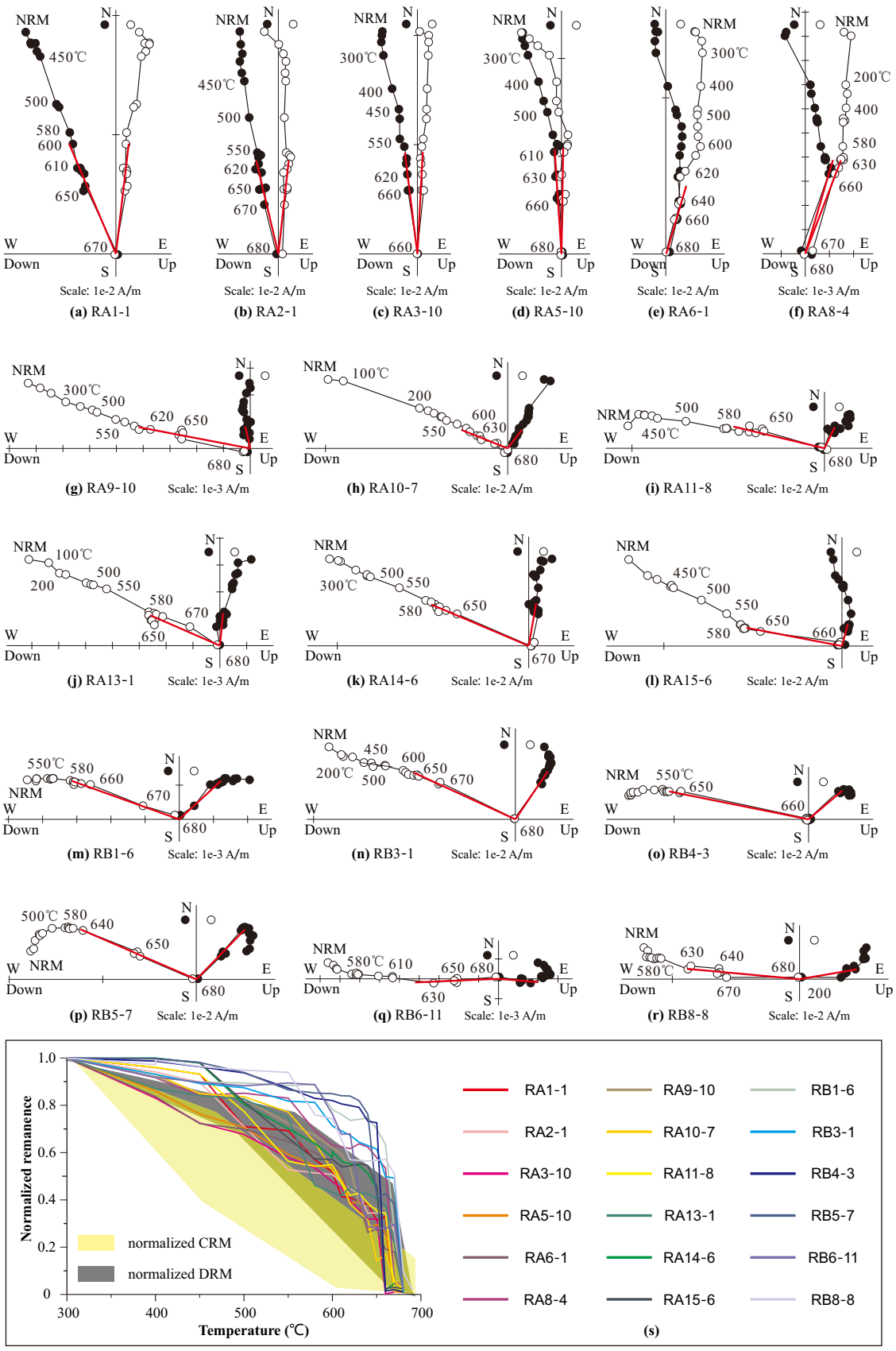
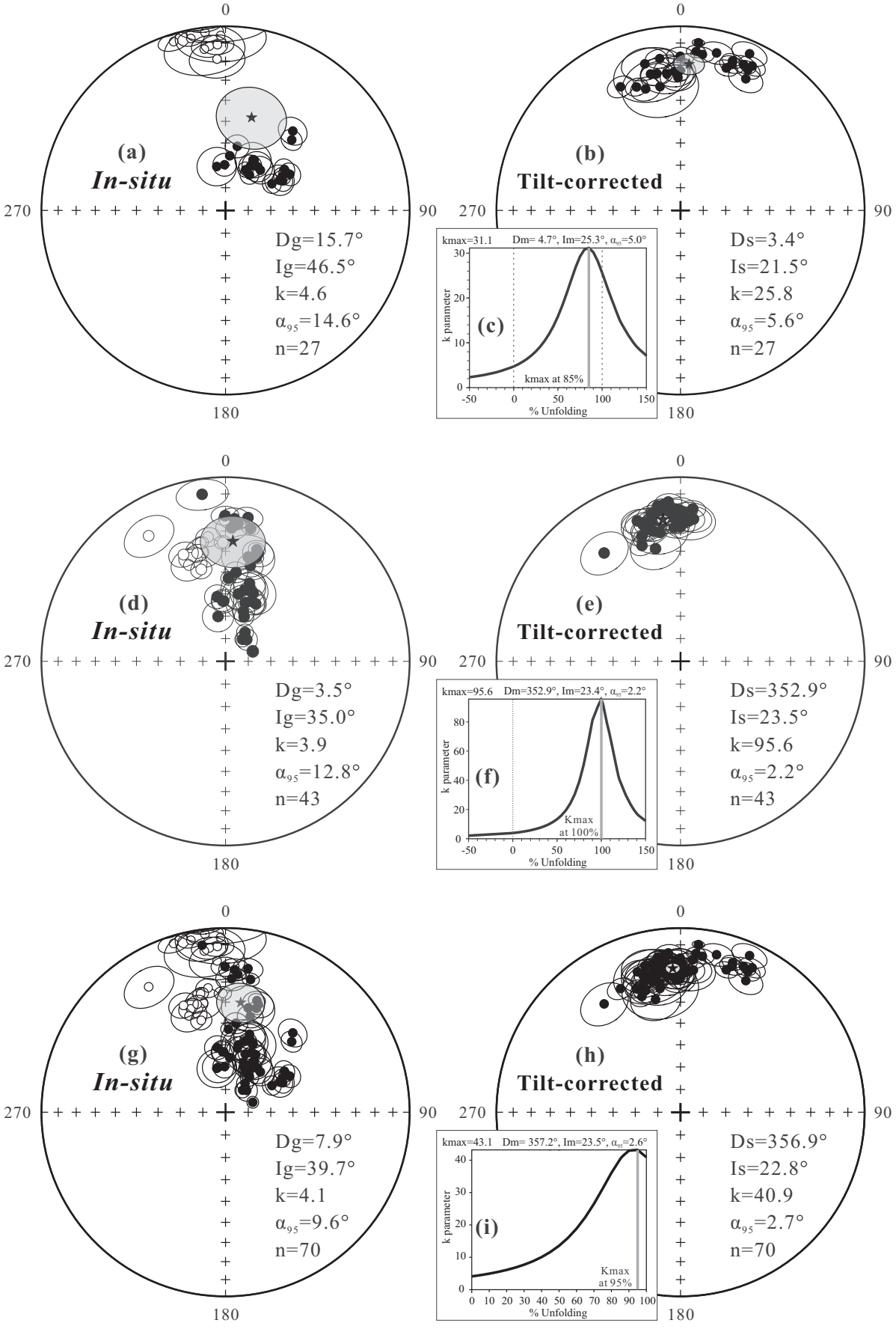


Fig. 5. (a-r) Stepwise thermal demagnetization curves of representative Shexing Fm redbeds in geographic coordinates. Stepwise thermal demagnetization curves are shown in vector component diagrams in which horizontal and vertical projections are combined into a single vector component diagram. Solid data points show vector endpoints projected onto the horizontal plane; open data points show vector endpoints projected onto the vertical plane; numbers adjacent to data points are demagnetization levels. (s) Normalized decay curve of the magnetization from 300 °C.



(caption on next page)

Fig. 6. Equal-area projections of site-mean HTC directions from this study (a, b) and those (d, e) from Tan et al. (2010) that have been reanalyzed, and that of the overall 70 sites from this study and Tan et al. (2010) (g, h). Results of the stepwise unfolding of the site-mean direction (c, f and i). Open circles are directions in the upper hemisphere; solid circles are directions in the lower hemisphere; the big ellipse (or circle), the α_{95} confidence limits; The stars with the gray area represent the mean of the HTC directions with surrounding stippled α_{95} confidence limits. n, number of sites used to calculate the mean; Dg, Ig (Ds, Is), declination and inclination before (after) tilt correction; k, precision parameter; α_{95}, the radius of the circle of 95% confidence; Dm and Im, declination and inclination.

Table 1
Site-mean directions of the HTC from the Shexing Fm redbeds, Linzhou area.

Site ID	n	Dg (°)	Ig (°)	Ds (°)	Is (°)	k	α_{95} (°)	Strike (°)	Dip (°)	Plat (°)	Plon (°)
RA1	8	342.9	−4.6	334.1	26.3	115.4	5.2	124	52	61.2	333.0
RA2	7	348.6	−5.3	340.4	29.6	185	4.4	121	49	67.2	327.5
RA3	8	353.4	−3	344	31.9	89.1	5.9	124	48	70.7	324.0
RA4	5	358.4	−5.6	350.6	32.4	27.2	14.9	124	48	75	308.2
RA5	9	354.4	−11.3	350.2	25.5	70	6.2	124	48	71.2	302.1
RA6	6	352.5	−12.7	347.6	25.1	23.9	14.0	129	55	69.7	308.3
RA7	9	357.2	−10.8	349.6	25.8	86.8	5.6	140	62	71.1	303.9
RA8	9	356.8	−18.8	354.3	26.4	27.5	10	124	56	73.2	290.7
RA9	7	348.7	70.2	346.3	19.2	50.5	8.6	255	51	66.2	306.6
RA10	7	29.4	67.5	359.9	17.4	147.3	5	253	56	69.0	271.5
RA11	8	36	66	358.9	28.6	70.3	6.7	245	47	75.3	275.4
RA12	7	28.7	64.5	357.5	25.4	59.1	7.9	245	47	73.3	279.7
RA13	8	10.4	61	352.2	17.9	116.5	5.2	245	47	68.0	292.2
RA14	7	4.4	65.7	359.8	25	999.9	1.8	266	41	73.2	271.9
RA15	8	358.8	69.8	358.3	25.8	121	5.1	268	44	73.6	277.1
RB1	9	60.1	66.3	17.8	19.9	147.8	4.2	265	58	64.3	227.3
RB2	7	26.3	69.9	5.6	14.4	312.7	3.4	265	58	66.8	257.0
RB3	8	29.2	64.4	6.1	9.3	705.2	2.1	260	60	64.1	257.2
RB4	7	34	71.1	8	15.6	215.9	4.1	266	59	66.8	250.7
RB5	7	40.9	66.4	12.9	13.3	99	6.1	266	59	63.9	241.0
RB6	8	60.8	62.5	26.4	21.3	72.3	6.6	271	52	59.0	213.2
RB7	9	60.7	57.1	26.4	13.5	111.5	4.9	266	59	56.1	218.8
RB8	8	59	60.5	19.2	15.1	213	3.8	261	61	61.4	228.3
RB9	9	62.7	61.4	19.4	17.1	132.9	4.5	261	61	62.1	226.6
RB10	6	55.2	58	23.4	14.8	210	4.6	265	56	58.7	222.0
RB11	7	43.2	46.4	25.6	16.4	170.7	4.6	261	40	57.8	217.9
RB12	8	40.3	42.9	23.8	7.8	79.8	6.2	258	47	55.6	225.7
Mean	206/27	15.7	46.5	3.4	21.5	25.8	5.6			71.1	262.0
										K = 26.6	A_{95} = 5.5

Notes: Site ID, site identification; n/N, number of samples used to calculate the mean; Dg, Ig (Ds, Is), declination and inclination before (after) tilt correction; k/K, precision parameter; α_{95}/A_{95}, radius of the circle of 95% confidence; Plon and Plat, longitude and latitude of VGP after tilt correction.

curves indicate that the Shexing Fm redbeds' remanence is DRM (Fig. 5s; Table S1).

Rocks may suffer from remagnetization due to strain reorientation of remanence. Six main stages of magnetic fabric development in progressive deformation have been recognized: undeformed condition, earliest deformation, pencil structure, weak cleavage, strong cleavage, and stretching lineation (Parés, 2004; Hrouda and Chadima, 2020). The shape parameter T of AMS results for section RA1–8 has a mean value of 0.063, revealing a dominance of neutral shape for this section, while sections RA9–15, RB1–5, and RB6–12 have mean values of 0.418, 0.258, and 0.507, respectively, suggesting a dominance of oblate shape for these sections. The minimum AMS principal axes are distributed in an approximately N-S orientation, implying that the deformed magnetic fabrics did not reach strong cleavage. The deformed magnetic fabrics of our sampling specimens should belong to stages 2 (earliest deformation) to 4 (weak cleavage), which is also consistent with their weak and moderate deformed magnetic fabrics (Weil and Yonkee, 2009).

Previous studies have shown that primary remanence can be isolated for specimens with pencil structure magnetic fabrics (Li et al., 2020a, 2020b). Cao et al. (2017) reported high-quality paleomagnetic results from redbeds and interbedded volcanics. The AMS results of the redbeds show primary sedimentary fabrics (Cao et al., 2017). Consistent inclinations for redbeds and interbedded volcanics were obtained, indicating no significant inclination shallowing for the redbeds. The consistent inclinations of the ChRM in redbeds between this study and the work by Cao et al. (2017) suggest that tectonic deformation of the AMS structure did not shallow or steepen inclinations.

The positive fold tests and stepwise unfolding results also support that the prefold remanence can be determined from specimens with weak cleavage magnetic fabrics (Fig. 6). Yi et al. (2015) pointed out that the strata in section B of Tan et al. (2010) did not overturn, which is different from the viewpoint of Tan et al. (2010). Therefore, Yi et al. (2015) proposed that the data of Tan et al. (2010) cannot be used for tectonic interpretation. Notably, the inclination of stratigraphic coordinates is the angle between the HTC direction and the strata bedding plane. There is a relation between inclination and paleolatitude, which is usually called the dipole formula. Therefore, no matter the strata overturn or not, the absolute value of the paleolatitude, as observed by Tan et al. (2010), will not change. Therefore, once we know the in-situ declination and inclination (Dg and Ig) and the bedding attitudes (Strike and Dip) of Tan et al. (2010), we can recalculate the tilt-corrected declination and inclination (Ds and Is). We have recalculated the site-mean directions for the previous results by tilt-correcting (see Table S2 in the supporting information). The site mean HTC directions are Dg = 3.5°, Ig = 35.0°, kg = 3.9, and α_{95} = 12.8° in situ and Ds = 352.9°, Is = 23.5°, ks = 95.6, and α_{95} = 2.2° after tilt correction (Fig. 6d and e; Table S2). The McFadden (1990) correlation fold tests yield $\xi_{1,g}$ ($\xi_{2,g}$) = 34.61 (40.19) in geographic coordinates and $\xi_{1,s}$ ($\xi_{2,s}$) = 0.47 (2.62) in stratigraphic coordinates, respectively. The critical values of the test statistic ξ are 7.63 and 10.79 at 95% and 99% confidence levels, respectively (McFadden, 1990), indicating a pre-fold origin. The stepwise unfolding of the mean direction yields a maximum grouping at 100% unfolding (Fig. 6f). These results show that the HTC of the redbeds from the Linzhou area is a pre-fold remanence and is most likely a

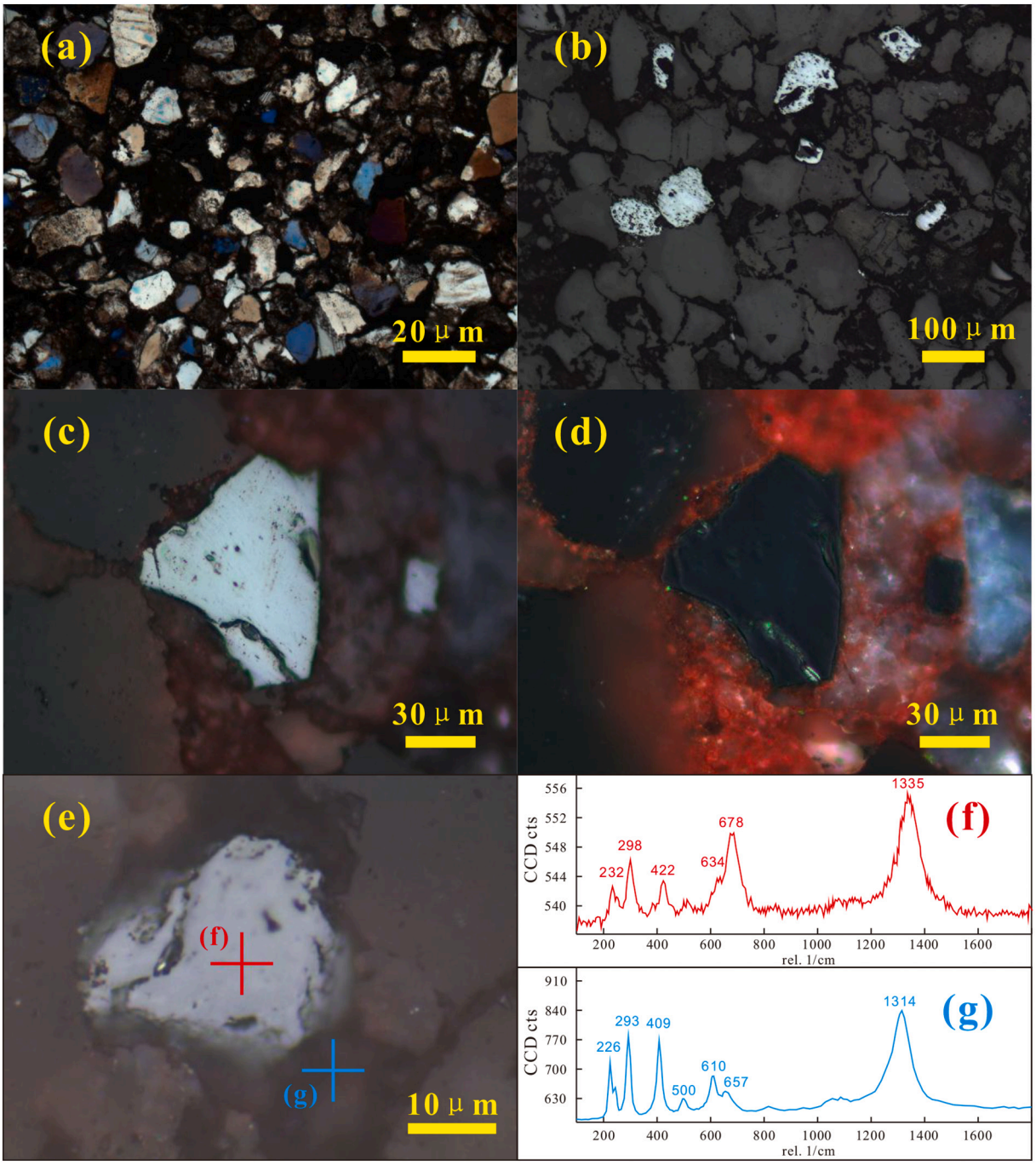


Fig. 7. Photomicrograph of redbeds and typical micro-spectroscopy analyses of detrital and pigmentary hematite. (a) Detrital minerals showing a clastic structure under crossed-polarized transmitted light; (b) Detrital hematite grains under plane-polarized reflected light; (c, d) Detrital hematite grains and surrounding pigmentary hematite under plane-polarized (c) and crossed-polarized (d) reflected light; (e, f) Raman micro-spectroscopy analysis of detrital and pigmentary minerals and corresponding results shown in the same color.

primary remanence.

Although the Shexing Formation redbeds have a depositional age of ~70–91 Ma, all the 70 sites were characterized by normal polarity, indicating that the sampling cores were not contained within a reversed polarity chron 33R (~79.9–83.6 Ma) (Gradstein et al., 2020). This phenomenon can be explained by the uneven distribution of our sampling sections, sites and cores due to different outcrop conditions (Fig. 1c and 2).

5.2. Have the HTC suffered from inclination shallowing?

As Fig. 6b shown, the site-mean HTC directions after till correction are distributed with east-west elongation, leading to an inclination shallowing with a flattening factor (f) of ~0.5. The E/I correction may produce a false appearance of significant inclination shallowing. For example, Tong et al. (2015) suggested that the E/I correction method may be influenced by the tectonic deformation of sampling strata, including vertical-axis rotations and plunging folds. Notably, local rotations are very common, especially in the orogenic belt like the Tibetan Plateau (e.g., Ma et al., 2014; Song et al., 2017; Yang et al., 2019). For

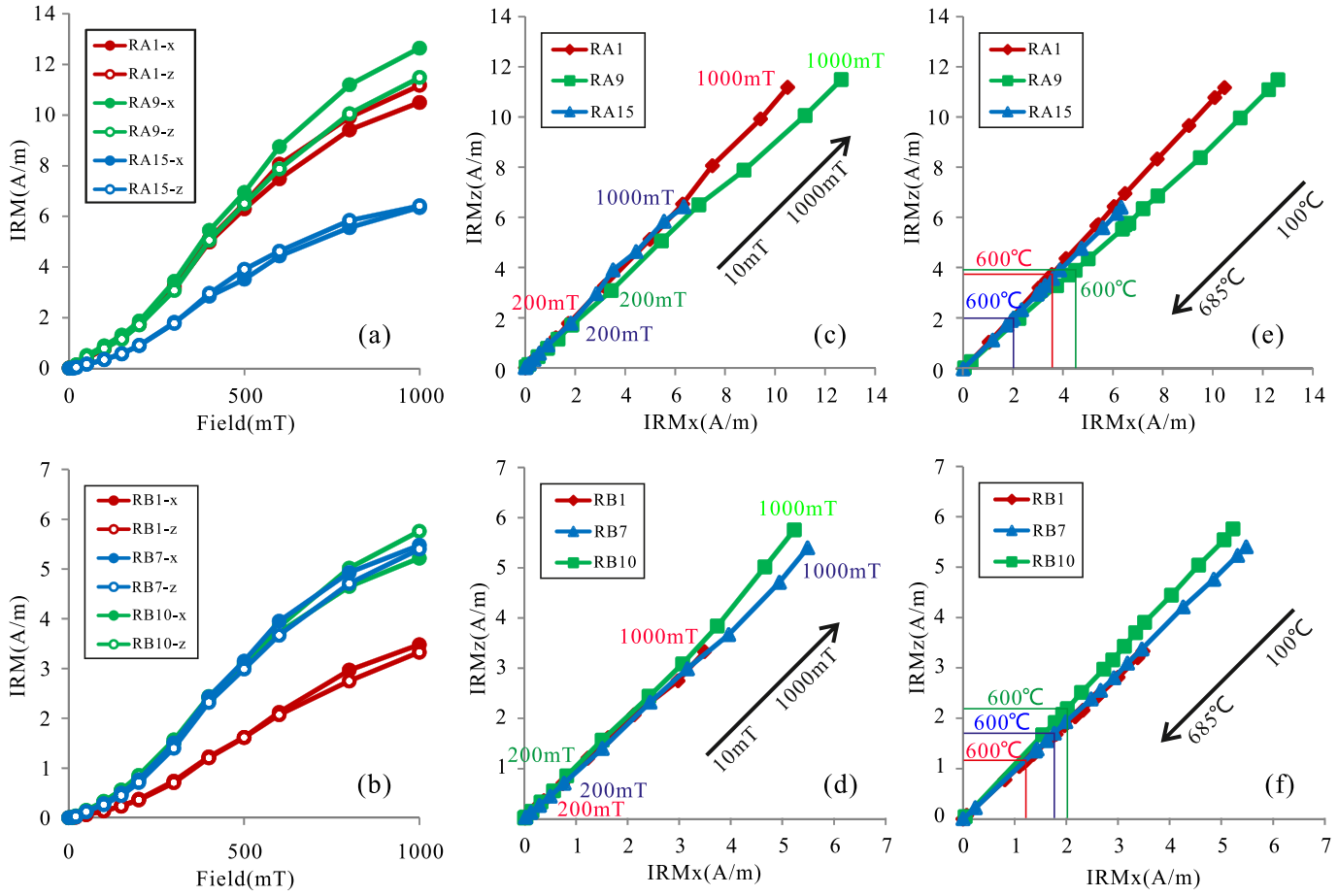


Fig. 8. (a, b) Plots of IRMx (parallel to bedding) and IRMz (perpendicular to bedding) acquisitions produced by applying a magnetic field at an angle of 45° to the bedding as a function of the increasing field strength; (c, d) The slope (IRMz/IRMx) of the least-squares-fit for the data points between 200 mT and 1000 mT used to estimate the magnetic anisotropy of hematite; (e, f) The slope of the thermal demagnetization of IRMz and IRMx between 580 °C and 680 °C used to estimate the inclination shallowing ratio for the inclination correction of the redbed samples.

example, the sites CQ21, 22 and CQ1–20 of Yang et al. (2015a) were collected from two sides of a river (fault), yielding similar inclinations but quite different declinations for the site-mean HTC directions. The different declinations probably result in an overcorrection of E/I analysis. Due to vertical-axis rotation or plunging fold (Tan et al., 2010), the bedding attitude of the sampling section of Linzhou Basin varies significantly, which makes the E/I correction invalid. Furthermore, volcanic rocks are widely accepted as being immune from shallowing effects (e.g., Ma et al., 2014). However, an E/I correction on our previous volcanic directions in section A by Ma et al. (2014) also yields a significant inclination shallowing. Therefore, it should be cautious to explain the E/I correction results, especially those from the orogenic belt.

Notably, the inclination shallowing correction method of Hodych and Buchan (1994) avoids the influence of unstable declination. Our results based on this method suggest that there is no inclination shallowing. Furthermore, previous studies found no significant difference between the Cretaceous volcanics' inclination and the redbeds' uncorrected inclination from Lhasa terrane (Bian et al., 2017, 2020; Chen et al., 2012; Ma et al., 2014; 2017; 2018; Sun et al., 2012; Tan et al., 2010; Yang et al., 2015a). This interpretation is also supported by the consistent inclinations between redbeds and interbedded volcanics in the adjacent area (Cao et al., 2017) in the central part of Lhasa terrane and the results on redbeds from the Dingqing area (Tong et al., 2017b). Our observed inclination is comparable with that of the Early Cretaceous volcanics from the stable Lhasa terrane (Li et al., 2017; Sun et al., 2008). Finally, no significant compactional signature in the redbeds can be found, indicating no significant inclination shallowing (Meng et al., 2018). Therefore, we conclude that no significant inclination shallowing is present in the Cretaceous redbeds from the central part of Lhasa terrane.

5.3. *Restoration of Asian southern margin before the India-Asia collision*

Previous studies had revealed that the Lhasa terrane was the southernmost terrane in the Late Cretaceous before India and Asia collided (Yin and Harrison, 2000; Ma et al., 2018). Because the present Lhasa terrane is extremely elongated in W-E direction and different parts of the southern margin of Lhasa terrane may have different paleolatitudes, to reconstruct the southern margin of Asia, two reference points RPW (31.7°N, 80.1°E) and RPC (29.3°N, 91.1°E) were chosen to calculate the paleolatitudes for the western and central parts, respectively (Fig. 1b; Table 2). The choice is based on the following reasons: (1) RPW and RPC are located on the Yarlung-Tsangpo suture zone; (2) the longitude of most investigations is located between ~80°E and ~ 91°E, especially many study areas are located at or near the Shiquanhe and Lhasa, at the longitude of ~80°E and ~ 91°E, respectively; (3) RPW and RPC span enough longitudinal distance (~11°) to shape the southern margin of Asia. The connecting line between RPW and RPC is in the orientation of ~NW286°.

To reconstruct the Asian southern margin, we have to figure out the paleolatitudes of the two reference points, RPW and RPC. Table 2 shows the Cretaceous paleomagnetic results from Lhasa terrane and the corresponding paleolatitudes for RPW and RPC. As paleomagnetic results

Table 2
Summary of Cretaceous paleomagnetic results from the Lhasa terrane.

ID	Lithology	Area	Slat (°N)	Slon (°E)	Age (Ma)	Plat (°N)	Plon (°E)	A_{95} (dp/dm) (°)	Paleolat (°N)	n/N	Criterion (Q)	References
Western part of the Lhasa terrane												
ZN	Volc	Cuoqin	31.4	85.1	~131–110	58.2	341.9	4.6	22.5 ± 4.6	162/18	123F5R7 (7)	Chen et al. (2012)
SQ	Volc+Lim	Shiquanhe	32.2	80.4	~116–113	69.1	319.8	4.8	19.8 ± 4.8	205/19	123F5□7 (6)	Bian et al. (2017)
QS	Volc	Yanhu	32.3	82.6	~132–120	61.4	192.9	2.1	17.7 ± 2.1	444/51	123F5D7 (7)	Ma et al. (2014)
DZ	Volc	Cuoqin	31.1	84.4	~121–117	70.5	292.9	7.4	14.9 ± 7.4	116/12	123F5D7 (7)	Yang et al. (2015a)
JG	Lim	Shiquanhe	32.2	80.8	~113–72	68.0	211.1	1.9	16.2 ± 1.9	274/38	123F5R7 (7)	Ma et al. (2018)
UC	Volc	Shiquanhe	32.4	80.1	~92.5	64.1	209.0	9.6	13.8 ± 9.6	78/10	123F5R7 (7)	Yi et al. (2015)
LZ	Volc	Cuoqin	30.6	85.2	~99–93	63.1	224.6	5.1	8.9 ± 5.1	112/14	123F5□7 (6)	Tang et al. (2013)
LD	Volc	Shiquanhe	32.4	80.1	~68	47.8	181.4	6.4	16.1 ± 6.4	300/17	123F5D7 (7)	Ma et al. (2017a, 2017b)
YR	Volc	Yare	31.6	82.2	~80	68.4	298.8	2.7	14.1 ± 2.7	136/15	123F5□7 (6)	Yi et al. (2015)
CQ	RB	Cuoqin	31.2	84.7	K_2	49.0	344.3	5.3	19.9 ± 5.3	291/33	123F5D7 (7)	Yang et al. (2015a)
GJ	RB	Geji	32.7	81.4	K_2	74.4	226.0	3.8	18.4 ± 3.8	431/54	123F5□7 (6)	Bian et al. (2020)
*CY	Lim	Shiquanhe	32.7	80.2	K	67.7	234.2	13.1/24.5	11.3 ± 13.1	22/3	1□3□5□7 (4)	Chen et al. (1993)
Mean paleolatitude calculated for the RPW in the Early Cretaceous									~18.7			This study
Mean paleolatitude calculated for the RPW in the Late Cretaceous									**~15.2**			**This study**
Mean paleolatitude calculated for the RPW in the Cretaceous									~16.6			This study
Central part of the Lhasa terrane												
DNF	Volc+Lim	Naqu	31.3	91.9	~120.2	66.9	281.2	6.1	6.5 ± 6.1	139/19	123F5D7 (7)	Li et al. (2017)
WR	Volc	Deqin	30.5	90.1	~114	66.4	220.3	6.9	13.2 ± 6.9	88/15	123□5□7 (5)	Sun et al. (2008)
CG	Volc+RB	Linzhou	29.9	91.1	75–68	70.5	269.6	4.9	9.8 ± 4.9	164/21	123□5R7 (6)	Cao et al. (2017a)
MX	Volc+RB	Maxiang	29.9	90.7	K_2	75	306.7	6.8	16.8 ± 6.8	126/20	123F5□7 (6)	Sun et al. (2012)
TY	RB	Dingqing	31.1	95.6	K_2	71.4	273.1	5.2	10.7 ± 5.2	150/15	123F5□7 (6)	Tong et al., 2017b
TX	RB	Linzhou	29.9	91.2	K_2	70.2	300.5	1.4/2.7	11.7 ± 1.4	377/43	123F5□7 (7)	Tan et al. (2010)
RR	RB	Linzhou	29.9	91.2	~91–70	71.1	262.0	5.5	10.6 ± 5.5	206/27	123F5□7 (7)	This study
RT	**RB**	**Linzhou**	**29.9**	**91.2**	**~91–70**	**71.8**	**281.5**	**2.6**	**11.4 ± 2.6**	**583/70**	**123F5□7 (7)**	Tan et al. (2010); **This study**
*NQ	Volc	Naqu	31.5	92.0	~96	78.0	282.0	4.0/6.9	17.5 ± 4.0	33/9	123□□□7 (4)	Lin and Watts (1988)
*QL	Volc	Chalicuo	31.7	91.0	~90	74.0	318.0	11.1/19.1	17.9 ± 11.1	20/4	1□3□□□□ (2)	Lin and Watts (1988)
*PJ	RB	Linzhou	29.9	91.2	K_2	68.0	340.0	6.7/11.6	19.6 ± 6.7	68/7	123F5D7 (7)	Pozzi et al. (1982)
*AN	RB	Barda	31.7	91.5	K_2	63.5	325.4	6.5	12.2 ± 6.5	49/6	123F5□7 (6)	Achache et al. (1984)
*AS	RB	Linzhou	29.9	91.1	K_2	71.2	288.4	7.9	11.3 ± 7.9	61/8	123F5□7 (6)	Achache et al. (1984)
*NM	RB	Nima	31.8	87.2	K_2	71.2	241.9	5.5/10.0	12.6 ± 5.5	59/8	123□5R7 (6)	Cao et al. (2017)
Mean paleolatitude calculated for the RPC in the Early Cretaceous									~9.8			This study
Mean paleolatitude calculated for the RPC in the Late Cretaceous									~11.9			This study
Mean paleolatitude calculated for the RPC in the Cretaceous									~11.3			This study

Notes: ID, paleopoles abbreviation used in the plot and text; Volc, volcanic rocks; Lim, limestone; RB, redbeds; K, Cretaceous; K_1, Early Cretaceous; K_2, Late Cretaceous; Slat and Slon, latitude and longitude of the sampling location; Plat and Plon, latitude and longitude of the pole; A_{95}, the radius that the mean pole lies within 95% confidence; dp/dm, semi-axes of elliptical error of the pole at a probability of 95%; paleolat, paleolatitude calculated for the reference points at RPW (31.7°N, 80.1°E) and RPC (29.3°N, 91.1°E) for the western and the central part of the Lhasa terrane; n/N, number of samples/sites used to calculate Fisher mean; Criteria (Q), data quality criteria (number of criteria met) after Van der Voo (1990) (1, well determined rock age; 2, sufficient sample number [$N > 24$, $k \geq 10$ and $\alpha_{95} \leq 16.0$]; 3, proper demagnetization techniques; 4, field tests; 5, structural control and tectonic coherence with the craton or block involved; 6, the presence of reversals; 7, no resemblance to paleopoles of younger ages [by more than a period]; F, positive fold test; R, positive reversal test; D, dual-polarity; "□", failed to meet this criterion). The data calculated in this study are shown with bold. Paleopoles signed with "*", calculated from less than ten sites, are neglect for further discussion.

become increasingly available, previous results with less than ten sites, which may not sufficiently sample the secular variation of the paleomagnetic field, are neglected for further discussion.

For the western part of the Lhasa terrane, a large number of paleomagnetic investigations have revealed that the Lhasa terrane maintained a stable paleolatitude of ~9–23°N at the time interval from ~132 to ~67 Ma (e.g., Bian et al., 2017, 2020; Ma et al., 2014, 2017a, 2017b, 2018; Tong et al., 2019; Yang et al., 2015a; Yi et al., 2015) (Table 2). The mean paleolatitudes calculated for the Early Cretaceous and Late Cretaceous eras are ~18.7°N at ~110–132 Ma (Chen et al., 2012; Bian et al., 2017; Ma et al., 2014; Yang et al., 2015a) and ~ 15.2°N at ~66–101 Ma (Tang et al., 2013; Yi et al., 2015; Ma et al., 2017a, 2017b; Yang et al., 2015a; Bian et al., 2020). These paleolatitudes are supported by a paleolatitude of 16.2 ± 1.9°N at ~72–113 Ma (Ma et al., 2018), yielding a stable paleolatitude (~16.6°N) for the western part of the Lhasa terrane during the Cretaceous (Table 2). Therefore, the RPW of the Lhasa terrane should be located at a paleolatitude of ~15.2°N in the Late Cretaceous.

For the central part of the Lhasa terrane, the mean paleolatitudes for the RPC are ~9.8°N in the Early Cretaceous (Li et al., 2017; Sun et al., 2008) and ~ 11.9°N in the Late Cretaceous (Cao et al., 2017; Sun et al., 2012; Tong et al., 2017b; Tan et al., 2010). Our results reveal that it is valid to use the uncorrected inclination of the redbeds' HTC directions from the Linzhou area for tectonic reconstruction. To obtain a reliable result from the Linzhou Basin, we recalculated the mean paleopole for 70 sites' virtual geomagnetic poles (VGPs) from our new results (Table 1) and those (Table S2) recalculated from Tan et al. (2010). The mean paleopole was located at 71.8°N, 281.5°E with A_{95} = 2.6°, yielding a paleolatitude of ~12.0 ± 2.6°N for the Linzhou area (29.9°N, 91.2°E) (Table 2). To test the influence of the *E*-W elongated distribution, we calculated and obtained the mean inclination of ~22.4 ± 1.7° for the 70 sites' redbeds using the inclination-only method (Arason and Levi, 2010), which corresponds to a paleolatitude of ~11.6 ± 1.0°N for the Linzhou area. The similar paleolatitudes calculated by two different methods indicate that the paleopole and corresponding paleolatitude of ~12.0 ± 2.6°N for the sampling area are valid for paleogeographic reconstruction (Table 2). Considering the negligible crust shortening within the Lhasa terrane since the Cenozoic (Murphy et al., 1997), the reference point RPC (29.3°N, 91.1°E) for the central part of the southern margin of Asia should be located at ~11.4°N at ~80 Ma. The observed paleolatitude (11.4° ± 2.6°N) for the reference point RPC is consistent with those of 10.7° ± 5.2°N, 9.8° ± 4.9°N and 16.8° ± 6.8°N calculated

from the data reported by Tong et al. (2017b), Cao et al. (2017) and Sun et al. (2012), respectively (Table 2).

A comparison of the Late Cretaceous paleolatitudes between the central and western reference points shows that the paleolatitude (~15.2°N) of the western part of Asia's southern margin was higher than that (~11.4°N) of the central part. Two points can determine a line, the shape of the central-western part of Asia's southern margin can, therefore, be roughly reconstructed into an orientation of ~NW 290° (Figs. 9 and 10a). Given the paleomagnetic uncertainty, the central-western part of Asia's southern margin in the Late Cretaceous was located at ~11–15°N along a WNW-ESE orientation (Figs. 9 and 10a).

The seismic tomography in a collision zone can be used to restore the paleogeography of pre-collisional Asia (Van der Voo et al., 1999). Because the Neo-Tethys slab remnant is found roughly under the paleoposition where it sank below the transition zone, Van der Voo et al. (1999) suggested that the Neo-Tethys slab descended more or less vertically into the deeper mantle, yielding the deeper-mantle high P-wave velocity anomaly II in Fig. 3b of Van der Voo et al. (1999). The present-day latitude of the deep slabs, i.e., the main remnant of the Neo-Tethyan oceanic lithosphere, can represent the approximate paleolatitudes of the Cretaceous subduction zone, i.e., the Southern Asian margin. The southern margin latitude of the anomaly is at ~16°N and ~10°N at 80°E and 90°E, respectively, which is highly consistent with our reconstruction for the southern margin of Asia, supporting a quasi-linear orientation for Asia before the India-Asia collision that is proposed by Yi et al. (2015). Moreover, the deeper-mantle high P-wave velocity anomaly suggests that the quasi-linear structure of Asia's southern margin can further extend in the ESE direction (Fig. 10a), between an orientation of ~310°E (Yi et al., 2015) and that of E-W direction (Ma et al., 2014).

The similarities of the magmatism, stratigraphy, paleobiogeography of the Tengchong and Lhasa terranes suggest that there has been a paleogeographical link between them since the Early Paleozoic (e.g., Shi et al., 2008; Xie et al., 2016; Zhang et al., 2013). The Shan-Thai block was suggested to be the eastern extension of the southern Qiangtang terrane (e.g., Metcalfe, 2011, 2013; Xie et al., 2016). The Baoshan terrane constitutes the northern part of the Shan-Thai block. Previous studies have shown that the Tengchong and Baoshan terranes have experienced a clockwise rotation of ~80° since the India-Asia collision (Kornfeld et al., 2014; Tong et al., 2016; Li et al., 2017, 2018a). The backward motion of "the accumulated ~80° of clockwise rotation during the Cenozoic" for the Tengchong and Baoshan terranes repositions them into the eastern extension of the Lhasa terrane and the Qiangtang terrane, respectively (Kornfeld et al., 2014; Tong et al., 2016; Li et al., 2018a). Therefore, the paleolatitudes of Lhasa and Tengchong terranes reconstructed Asia's southern margin in a WNW-ESE orientation (Fig. 10a).

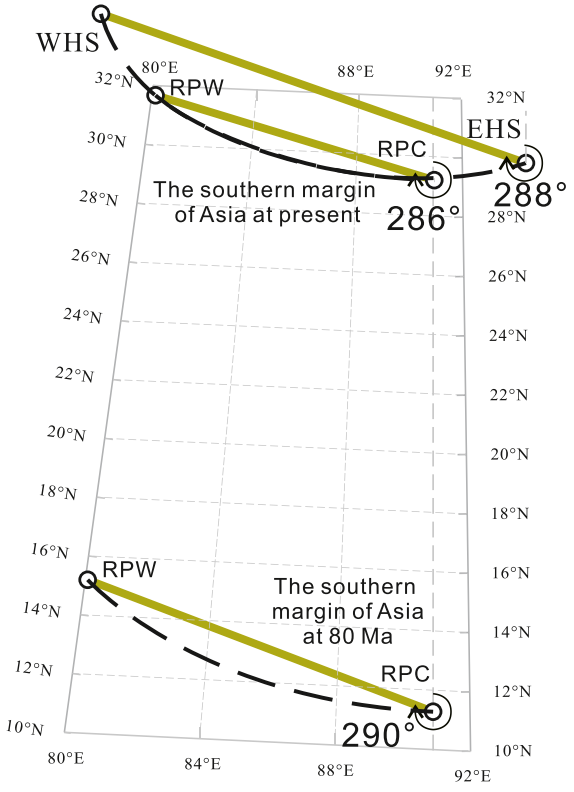


Fig. 9. Reconstruction of the southern margin of Asia based on the (paleo-) latitudinal constraint on the reference points RPW (31.7°N, 80.1°E) and RPC (29.3°N, 91.1°E) at the Late Cretaceous and present. EHS, eastern Himalayan syntaxis; WHS, western Himalayan syntaxis.

5.4. Crustal convergence within Asia north of the Indus-Tsangpo suture zone

The orientation of ~NW 290° is similar to that of NW286° for the connecting line between RPW and RPC and to that of ~NW 288° for the connecting line between the eastern and western Himalayan syntaxis. The parallel orientation between the present-day connecting line between RPW and RPC and their paleo-positions suggests that the northward movement of these two reference points is similar since the Late Cretaceous.

Compared with the stable expected paleolatitudes for the reference points (Torsvik et al., 2012), the similar northward movement of Lhasa terrane's reference points implies similar crustal convergence (Ma et al., 2017a, 2017b). The India-Asia collision led to a significant crustal convergence, including crustal shortening and thickening accommodated by the Cenozoic fold and thrust belts within Asia since the collision (e.g., van Hinsbergen et al., 2011) and the lateral extrusion of crustal materials around the southeastern edge of the Tibetan Plateau since the Oligocene (e.g., Tong et al., 2017a). The difference between the observed paleolatitude and the expected one calculated from the reference pole is the latitudinal convergence within Asia north of the study area. The observed paleolatitudes for the reference point RPW (31.7°N, 80.1°E) and RPC (29.3°N, 91.1°E) are ~15.2°N and ~ 11.4°N, respectively, while the expected paleolatitudes calculated from the reference pole of East Asia at 80 Ma are 23.8° ± 3.4°N and 22.8° ± 3.4°N, respectively (Cogné et al., 2013). The paleolatitude differences between the paleolatitude observed at 80 Ma and that expected at 80 Ma are ~8.6° and ~ 11.4° for the reference point (31.7°N, 80.1°E) and (29.3°N, 91.1°E), respectively. These results reveal that the latitudinal convergences are ~1000 km and ~ 1300 km, respectively, for the western and central parts within Asia since 80 Ma. Therefore, there was no significant crustal convergence difference within Asia north of the Indus-Tsangpo suture zone between the western and central part of the Tibetan Plateau at the paleomagnetic confidence level (Ma et al., 2017a, 2017b).

The amount of crustal convergence is consistent with that of crustal shortening of ~1000 km absorbed by the Cenozoic thrust and fold belts within Asia (Ma et al., 2014, 2017b; Yin and Harrison, 2000; van Hinsbergen et al., 2011), indicating that most of the crustal convergence is crustal shortening. The difference between the most probable amounts of the latitudinal convergence of the central and western parts may be explained by the lateral extrusions of crustal material, which plays a more critical role in the central and eastern parts of the Tibetan Plateau than those in the western part (Otofuji et al., 2007; Tong et al., 2016, 2017a).

Notably, the expected paleolatitudes for the reference points RPW (31.7°N, 80.1°E) and RPC (29.3°N, 91.1°E) calculated from the reference pole of East Asia at 50 Ma are 23.8° ± 3.4°N and 22.1° ± 3.4°N, respectively (Cogné et al., 2013). These paleolatitudes are strikingly similar to that at 80 Ma, indicating that East Asia remained relatively

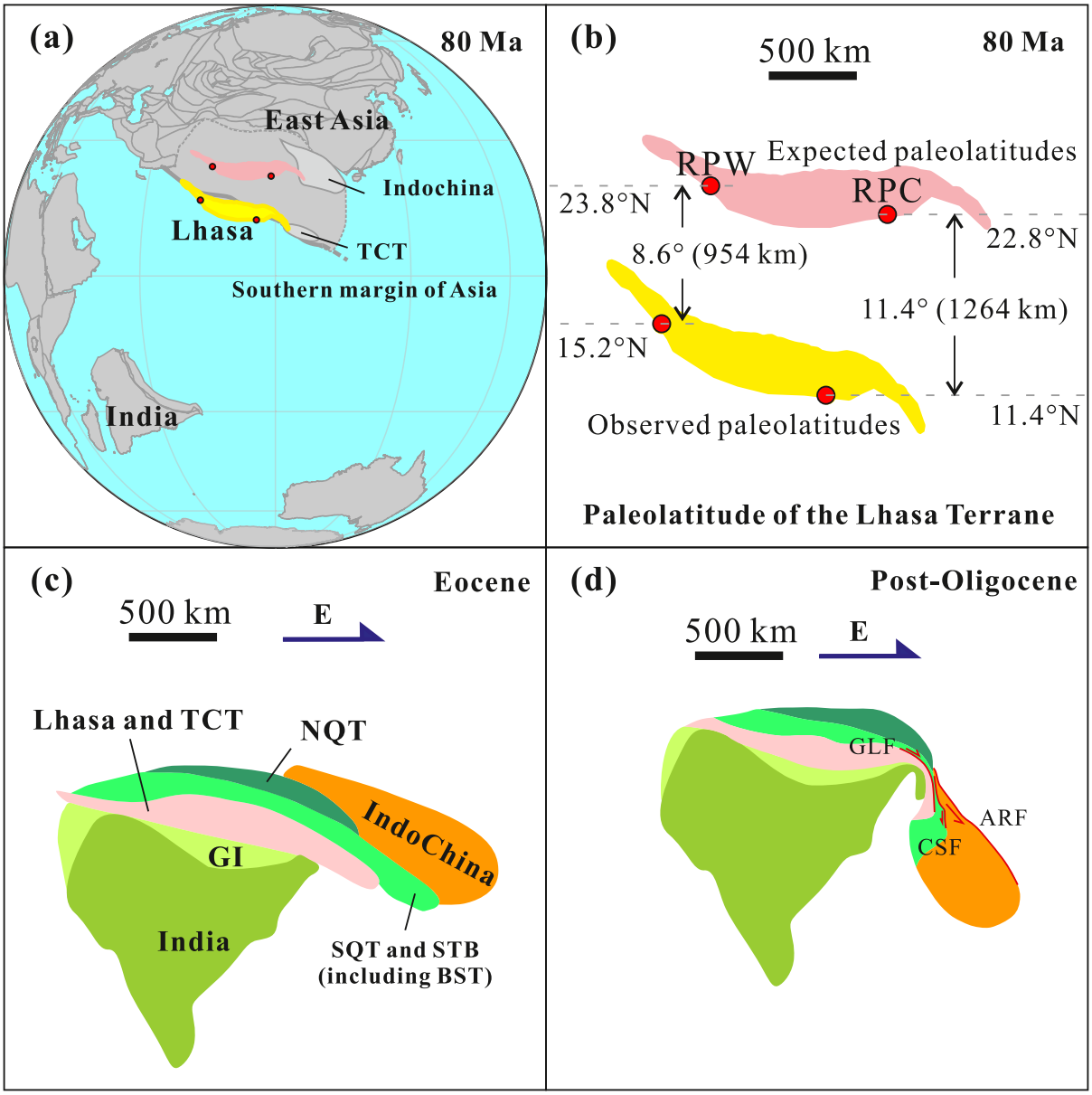


Fig. 10. (a) Paleogeography of the southern margin of Asia at ~80 Ma; (b) the crustal convergence within the Asia continent since 80 Ma; (c,d) the Cenozoic crustal movement characteristic of the southeastern part of Tibetan Plateau, modified after Tong et al. (2017a). The paleogeographic reconstructions were performed using Gplates. The radius of the Earth is 6371 km. The paleolatitudes of the Lhasa Terrane at ~80 Ma were determined by this study. The Asian southern margin was determined by Van der Voo et al. (1999). The restoration of TCT to the eastern ward of the Lhasa Terrane was based on Xie et al. (2016) and Kornfeld et al. (2014). The post-Cretaceous southward movement of the Indochina was determined by Yan et al. (2017). The gray areas in panels (a) are (inferred) continents, and the blue areas are oceans. The yellow terranes in panels (b) were reconstructed using the observed paleolatitudes, while the pink terranes show the expected paleolatitudes. The paleolatitude difference between observed paleolatitude from the study area and the expected one calculated from the continent's reference pole can be used to constrain the amount of crustal convergence. Abbreviations: BST, Baoshan terrane; NQT, Northern Qiangtang terrane; SQT, Southern Qiangtang terrane; STB, Shan-Thai block; TCT, Tengchong terrane; GLF, Gaoligong strike-slip fault; CSF, Chongshan strike-slip fault, ARF, Red-river strike-slip fault. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

stable at the time interval of ~80–50 Ma. This conclusion is highly consistent with our previous conclusion that Asia's pre-collisional southern margin is stable before India's indentation into Asia (e.g., Ma et al., 2014; Sun et al., 2012).

5.5. *The influence on the shape of the Himalayan-Tibetan orogen based on our reconstruction*

On the Asian side, previous studies have revealed that significant lateral extrusions have occurred on the southeastern edge of the Tibetan Plateau because of the continuous compression between India and Asia during the Cenozoic (e.g., Tong et al., 2017a; Yan et al., 2017). The southward extrusion of crustal materials near the EHS seems to fit well with the present-day global positioning system (GPS) data (Gan et al., 2007). Notably, the GPS data provide no information on the evolution of the India-Asia orogenic system at million-year scales (Haproff et al., 2018), especially for the crustal deformation caused by the collision and subsequent indentation of India into Asia during the last ~50 Myr. However, the restoration of Asia's southern margin before the India-Asia collision can provide us with an excellent opportunity to understand the crustal deformation caused by different crustal movement patterns of microcontinents within Asia.

The Tengchong terrane and the Baoshan terrane are the eastern extensions of the Lhasa and Qiangtang terranes, respectively, during the Late Cretaceous (Fig. 1a; Tong et al., 2016, 2017a). Based on our reconstruction of Asia's southern margin, the Tengchong terrane should be located at a similar paleolatitude to that of the Lhasa terrane, thereby suggesting a northward movement of the Lhasa-Tengchong terrane during the Cenozoic (Fig. 10c,d). This interpretation is supported by the northward movement of Baoshan terrane, located on the east of Tengchong terrane in present-day coordinates. The Baoshan terrane moved from 9.1° ± 5.6°N at the Paleocene, to 19.4° ±2.5°N at the Miocene, to 24.3°N at present (Tong et al., 2016). The paleolatitude evolution of the Tengchong and Baoshan terranes suggest that the Tengchong terrane, which is closer to the EHS, moved northward faster than the Baoshan terrane. Besides, Asia's southern margin makes a sharp bending around the EHS, implying that the Tengchong terrane moved northward slower than the Lhasa terrane (Fig. 10c,d).

The latitudinal motions of the crustal material show a remarkable trend. The northward motion decrease from Lhasa to Tengchong terranes, then to Baoshan terrane, and finally to Indochina, which moved southward with a clockwise rotation (Li et al., 2017; Yan et al., 2017). This trend may be caused by the lack of the direct push from India to the east of the EHS, the northward movement with clockwise rotation, which is stemmed by the slower Asia landmass, and the southeastward extrusion of the Indochina along the Ailaoshan-Red River Fault. Accompanied with the indentation of India into Asia, the Tibetan Plateau and its southeastern margin have undergone different latitudinal movements as manifested by different clockwise rotations around the EHS and lateral extrusions along the large-scale strike-slip faults, including Gaoligong strike-slip fault (GLF), Chongshan strike-slip fault (CSF), Red-river strike-slip fault (ARF) (Fig. 10d).

On the Indian side, the shape of the northern margin of Greater India remains unclear due to a lack of reliable evidence. Can the pre-collisional southern margin of Asia make a difference in shaping the Himalayan orogen? Based on (paleo-) latitudinal constrain from the Lhasa terrane, the southern margin of Asia can be reconstructed into a relatively linear structure with ~NW 290° at ~80 Ma and ~ NW 286° at present, respectively (Fig. 9). With India's indentation into Asia, Asia moved northward significantly (~16.5–17.9°) (Fig. 9). The shape of the southern margin of Asia did not change significantly due to the collision. Besides, the similar orientation between the southern margin of the Asia continent in the Late Cretaceous and that of the connecting line between the eastern and western Himalayan syntaxis indicates that the shape of the suture zone depends mainly on the Asia continent, but not on the shape of Greater India which is still under intense debate (Ali and Aitchison, 2005; Meng et al., 2019, 2020; Yang et al., 2015b, 2019; Yi et al., 2011; Yuan et al., 2020). Therefore, we conclude that the shape of Asia's southern margin plays a critical role in shaping the Himalayan orogen.

6. Conclusions

New paleomagnetic data for the Late Cretaceous Shexing Fm redbeds pass fold tests, indicating a pre-fold origin. These data are immune from inclination shallowing and can thus be used for tectonic reconstruction. Paleomagnetic results from the redbeds of 70 sites reveal that Asia's southern margin was at a paleolatitude of ~11.4°N for the reference point RPC (29.3°N, 91.1°E). Our new results, combined with the seismic tomography and paleomagnetic data from the Lhasa, Tengchong and Baoshan terranes, suggest that the southern margin of Asia should be reconstructed into an orientation of WNW-ESE. The amount (~1264 km) of latitudinal convergence in the central part of the Tibetan Plateau within Asia since 80 Ma has been accommodated by considerable crustal shortening and minor lateral extrusion within Asia since the India-Asia collision. Different latitudinal movements of the Lhasa, Tengchong, Baoshan and Indochina terranes are accompanied by clockwise rotations around the EHS and lateral extrusions along the large-scale strike-slip faults. This conclusion is consistent with the sharp bending around the EHS of the Tibetan Plateau. Therefore, the shape of the southern margin of Asia plays a critical role in shaping the Himalayan orogen.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.palaeo.2021.110821.

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